

Multiple Choice (10 marks)

1. By definition, the electric displacement current through a surface S is proportional to the
 - (a) magnetic flux through S
 - (b) rate of change of the magnetic flux through S
 - (c) time integral of the magnetic flux through S
 - (d) rate of change of the electric flux through S
2. A refracting telescope consists of two converging lenses separated by 100 cm. The eyepiece lens has a focal length of 20 cm. The angular magnification of the telescope is
 - (a) 4
 - (b) 5
 - (c) 6
 - (d) 20
3. Which of the following is true about any system that undergoes a reversible thermodynamic process?
 - (a) There are no changes in the internal energy of the system.
 - (b) The temperature of the system remains constant during the process.
 - (c) The entropy of the system and its environment remains unchanged.
 - (d) The entropy of the system and its environment must increase.
4. Consider an electron in the $n = 4$, $l = 1$ state in hydrogen. Which of the following final states can NOT be reached by an allowed transition?
 - (a) $n = 3$, $l = 2$
 - (b) $n = 3$, $l = 1$
 - (c) $n = 3$, $l = 0$
 - (d) $n = 2$, $l = 0$
5. A single-electron atom has the electron in the $l = 2$ state. The number of allowed values of the quantum number m_l is
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 5

True / False (10 marks)

Answer True or False

6. True or False: A grating spectrometer can achieve a resolving power of 10,000 when resolving two wavelengths close together.
7. True or False: In a single-electron atom with $l = 2$, there are 5 allowed values for the magnetic quantum number m_l .
8. True or False: In superconductors, the attraction between Cooper pairs arises from their interactions with the ionic lattice of the material.
9. True or False: The specific heat of a doped semiconductor can reveal the sign of its charge carriers.
10. True or False: If the absolute temperature of a blackbody is increased by a factor of 3, the energy radiated per second per unit area increases by a factor of 81.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of charge redistribution that occurs when a charged capacitor is connected to two identical capacitors in series. What is the final voltage across the initially charged capacitor, and explain your reasoning.
12. Discuss the implications of length contraction in special relativity by calculating the speed at which an observer must travel parallel to a 1.00 m rod to perceive its length as 0.80 m.

End of Paper